

Attitude adjustment: Trends in the semantics of clausal-embedding

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Day 1



joint work with Tom Roberts, captured capturing an audience
roto.m.github.io/

Outline

Introduction

Semantics 101 and the problem with embedded clauses

Properties of clauses embedded under 'believe'

This course is about **attitude reports**, sentences like:

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So can the **complementizer**: *that, whether, if, to, ...*

the **matrix subject**: *people, cats, signs (?)*, ...



"The sign says stop."

One syntactic property:

These are sentences in which a predicate *subordinates* another sentence.

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We'll call these 'attitude' or 'clause-embedding' predicates.

And the clauses themselves, 'embedded', 'complement', or 'subordinate' clauses.

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The following, e.g., are not instances of subordination.

- (2) a. David is happy. Jesse stayed.
- b. David is happy because Jesse stayed.

One conceptual property:

These and similar sentences describe '**something mental**', which we call an attitude.

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Attitude *reports*... because they report on (other) people's attitudes.

Historical side note

Bertrand Russell seems to be credited for the word 'attitude' to describe psychological states and he discusses what to call these verbs in the passage below (in *The Philosophy of Logical Atomism*, sec. 4.4).

4. The question of nomenclature.

What sort of name shall we give to verbs like “believe” and “wish” and so forth? I should be inclined to call them “propositional verbs”. This is merely a suggested name for convenience, because they are verbs which have the *form* of relating an object to a proposition. As I have been explaining, that is not what they really do, but it is convenient to call them propositional verbs. Of course you might call them “attitudes”, but I should not like that because it is a psychological term, and although all the instances in our experience are psychological, there is no reason to suppose that all the verbs I am talking of are psychological. There is never any reason to suppose that sort of thing. One should always remember Spinoza’s infinite attributes

Can you think of any verbs that subordinate clauses but that aren't 'mental' or 'psychological'?

We'll dwell on the notion of a 'proposition' for a moment.

- (1) a. Chloe thinks that Brno is in Austria.
- b. David is happy that Jesse stayed.

Propositions have always (roughly) been associated with the meaning of **declarative sentences**, we'll return to this shortly.

So in calling his set of verbs 'propositional verbs', Russell is restricting his attention to declarative-taking verbs.

But our set of verbs take all sorts of complements.

Interrogative complements are fairly well-studied.

- (3) c. Theo wondered if she should buy cheese.
asked, is curious, ...

Imperatives a little bit as well:

- (4) John_i said call his_i mom. Crnič & Trinh (2011)

so our empirical landscape is very broad, that's ok

(Why) are attitude reports interesting?

recursion?

- (5) a. I know that Chloe thinks that Brno is in Austria.
- b. My friend is curious if I know that Chloe thinks that Brno is in Austria.

...

a universal property of human languages?

(Why) are attitude reports interesting?

because mental states themselves are important, then so might the sentences that we use about them—belief, knowledge, desire, ...

(Why) are attitude reports interesting?

attitude reports have interesting semantic properties whose study advances linguistic theory

- (6) a. A unicorn is in the garden.
- b. Barbara thinks that a unicorn is in the garden.

What, if anything, are we talking about when we use the words ‘a unicorn’, in (6b)?

And how do we model that, as different from the use in (6a)?

(attitude reports are not alone in having these properties of course)

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This is important for descriptive and theoretical purposes, but also for broader cognitive scientific inquiry.

Searching for a universal constraint on the possible denotations of clause-embedding predicates*

Floris Roelofsen
University of Amsterdam

Wataru Uegaki
University of Edinburgh

Abstract We propose a new cross-linguistic constraint on the relationship between the meaning of a clause-embedding predicate when it takes an *interrogative complement* and its meaning when it takes a *declarative complement*. According to this proposal, every clause-embedding predicate V satisfies a constraint that we refer to as P-TO-Q ENTAILMENT. That is, for any exhaustivity-neutral interrogative complement Q , if there is an answer p to Q such that $\lceil x \text{ Vs } p \rceil$, then it follows that $\lceil x \text{ Vs } Q \rceil$. We discuss empirical advantages of this proposal over existing proposals and explore potential counterexamples to P-TO-Q ENTAILMENT.

Recent large scale projects investigating attitude verbs

MECORE

Edinburgh & Konstanz

many semantic and 'combinatorial' properties of many attitude verbs in many languages

<https://wuegaki.ppls.ed.ac.uk/mecore/>

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the morphosyntax of complement clauses and how they get integrated into the matrix clause, also cross-linguistic

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The MegaAttitude Project

Johns Hopkins & Rochester

many semantic and syntactic properties of ALL the attitude verbs in English

<https://megaattitude.io/>

You too can join the excitement: the MECORE and MegaAttitude datasets are public.

An updated version is upcoming for MECORE.

IMMAGES is still in the process of collecting data.

More specific focus of the course

Now, given that you're here...

This course centers around **two questions**:

How should we represent the meanings of attitudes in a formal semantic system?

How does this meaning interact with other parts of grammar?

A living roadmap for the week

Day 1: Background, terminology and description

Day 2: A modern semantics for attitude reports: Hintikka

Day 3: A refinement of Hintikka, probably no farewell:
Incorporating event semantics into the mix

Day 4: A special topic applying these notions and/or an
exercise

Day 5: TBD

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Formal semantics abridged abridged

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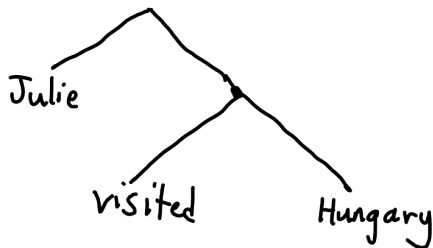
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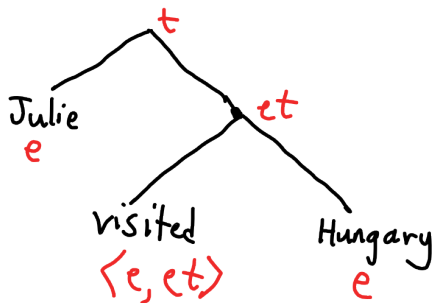
Meaning of a sentence is of type t

Composition occurs via **Function Application** of two sisters in a tree (requires binary branching)

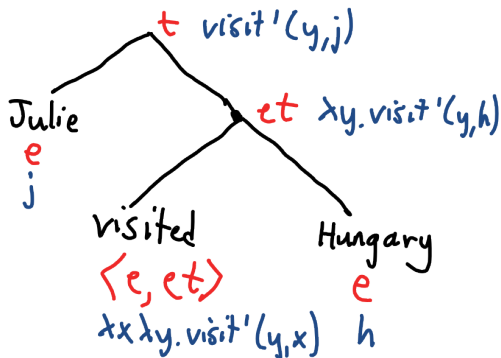
Compositionality example



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(different notation from H&K for convenience)

The nesting problem

So far so good..?

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Let's look at attitude reports, which have clauses as proper parts.

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- >look inside
- >another sentence



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How to combine **clausal meanings** with **attitude meanings**?

First, let's illustrate the puzzle.

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their **truth conditions**

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- a. Boris is a fluffy kitten.
 - b. $\llbracket (7a) \rrbracket = 1$ iff Boris is a fluffy kitten
 - c. $\llbracket (7a) \rrbracket = \llbracket \text{Boris} \rrbracket \in \llbracket \text{kitten} \rrbracket \wedge \llbracket \text{Boris} \rrbracket \in \llbracket \text{fluffy} \rrbracket$

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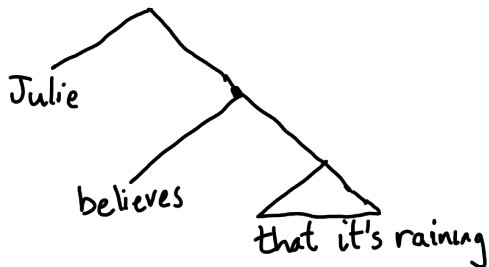
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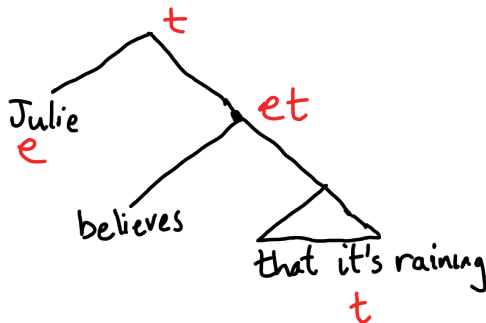
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- (8) Deniz thinks that Boris is a fluffy kitten.
- ✗ Boris is a fluffy kitten.
 - Deniz has some mental relation to the sentence 'Boris is a fluffy kitten'
 - ✗ Deniz has some mental relation to the truth value 'true'

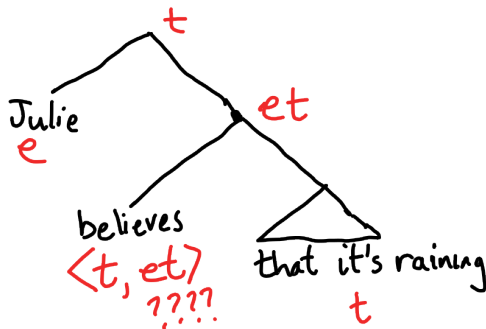
The nesting problem, tree-itized



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Observation

Upshot: The meaning of embedded clauses in attitude reports is not straightforwardly of type t .

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There is some additional evidence that *embedded* clauses have some special meaning properties that unembedded clauses lack....

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Substitution ('Frege's puzzle')

Do these two sentences mean the same thing?

- (9)
- a. Péter Magyar is riding a unicycle on the roof.
 - b. The Prime Minister of Hungary is riding a unicycle on the roof.

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What about these two?

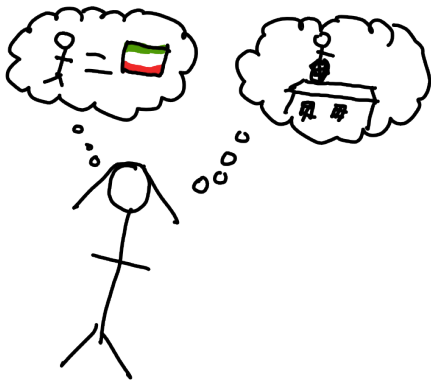
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- a. Mafalda believes that Péter Magyar is riding a unicycle on the roof.
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Depicting the difference

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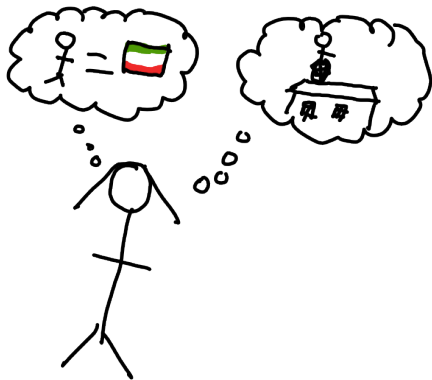
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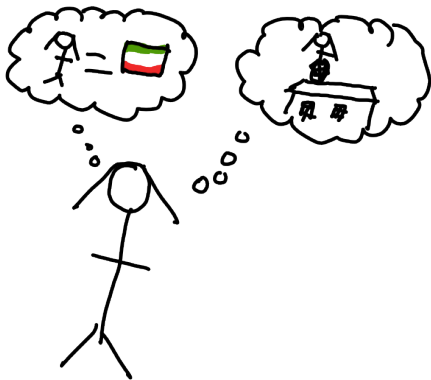
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- (10) a. Mafalda believes that Péter Magyar is riding a unicycle on the roof. **True**
- b. Mafalda believes that the PM of Hungary is riding a unicycle on the roof. **True**



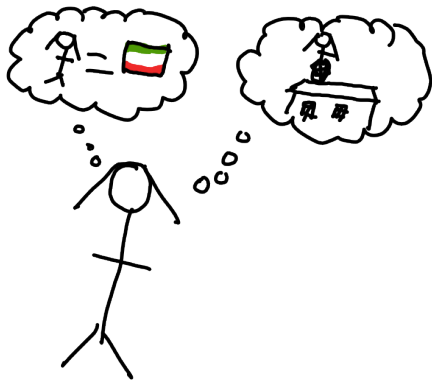
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Depicting the difference

- (10) a. Mafalda believes that Péter Magyar is riding a unicycle on the roof. **False**
- b. Mafalda believes that the PM of Hungary is riding a unicycle on the roof. **True**



Empty predicates



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- (11)
- a. Jackalopes are fierce.
 - b. Mafalda believes that jackalopes are fierce.
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(11a): Commits me to the existence of jackalopes

(11b)/(11c): Commits Mafalda (**but not me**) to the existence of jackalopes

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Tomorrow: A theory for the semantics of attitudes that accounts for these facts—and are all attitudes the same?